

Question:

What is the de Broglie wavelength of an electron with a kinetic energy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>
Given the de Broglie wavelength formula $\lambda = h/p$, where λ is the wavelength, h is Planck's constant (6.626×10^{-34} Js), and p is the momentum of the particle. For an electron with a kinetic energy of 1 eV, we need to calculate its momentum. Using the kinetic energy formula $E = \frac{1}{2}mv^2$ and the standard formula for kinetic energy $E = \frac{1}{2}mv^2$, we can find the velocity v and then the momentum $p = mv$. The mass of an electron is $m = 9.11 \times 10^{-31}$ kg. The energy 1 eV is 1.6×10^{-19} J. Solving for v gives $v = \sqrt{2E/m} \approx 5.93 \times 10^6$ m/s. Then $p = mv \approx 5.4 \times 10^{-24}$ kg m/s. Finally, $\lambda = h/p \approx 1.23 \times 10^{-7}$ m = 1.23×10^{-5} cm. Answer: G.

